EC39001: Analog Communication Lab
Experiment 1: Basic Filter Design

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1 Introduction

The objective of this experiment is to design and implement basic filters used in analog communication systems. The experiment involves the design of 1st order passive low-pass, high-pass, and band-pass filters, as well as an active low-pass filter using an operational amplifier. Additionally, we study the behavior of these filters by determining their 3dB frequencies and measuring the signal amplitudes at specific frequencies.And at last we build an Adder Circuit using Op-Amp .

2 Key Objectives

1. To design and analyze a 1st order passive low-pass filter (LPF) and high-pass filter (HPF).
2. To design and analyze a band-pass filter using the above LPF and HPF.
3. To implement a 1st order active LPF using an Op-Amp with a gain of 2.
4. To measure and verify the 3dB frequencies of the designed filters.
5. To measure the amplitude of the signal at $f_L/2$ and $2f_H$.
6. To add two voltage signals using an Op-Amp based adder circuit.

3 Components Used

- Resistors (various values as per design)
- Capacitors (various values as per design)

- Operational Amplifier (Op-Amp)
- Function Generator
- Digital Storage Oscilloscope (DSO)
- Breadboard and connecting wires
- DC Power Supply

4 Brief Theory

Filters are used to allow or block signals based on their frequency components:

- **Low-Pass Filter (LPF):** Allows frequencies below the cutoff frequency f_H to pass and attenuates higher frequencies. The cutoff frequency is given by:

$$f_H = \frac{1}{2\pi RC}$$

- **High-Pass Filter (HPF):** Allows frequencies above the cutoff frequency f_L to pass and attenuates lower frequencies.

$$f_L = \frac{1}{2\pi RC}$$

- **Band-Pass Filter (BPF):** Formed by Cascading LPF and HPF, passing frequencies between f_L and f_H , where f_L and f_H are cut-off frequencies of respective LPF and HPF .

- **Active LPF:** Implemented using an Op-Amp to provide amplification. The gain is:

$$\text{Gain} = 1 + \frac{R_f}{R_2}$$

- **Adder Circuit:** An Op-Amp based summing circuit is used to add two input signals according to:

$$V_{\text{out}} = (-1)^G \left[x_1(t) + \big((G-1) \bmod 4 + 1\big) x_2(t)\right]$$

5 Circuit Design and Circuit Diagram

5.1 Low-Pass Filter (Passive)

- Cutoff frequency: $f_H = 4[(G-1) \bmod 4] + 10$ kHz
  $f_H = 4[(6-1) \bmod 4] + 10$ kHz
  $f_H = 4[5 \bmod 4] + 10$ kHz
  $f_H = 4 * 1 + 10$ kHz
  $f_H = 14$ kHz

- Calculation:
  We have to taken $C = 8.36$ nF , Closest value of $R = 1.18$ kΩ , Corresponding Cutoff frequency 16.12 kHz .

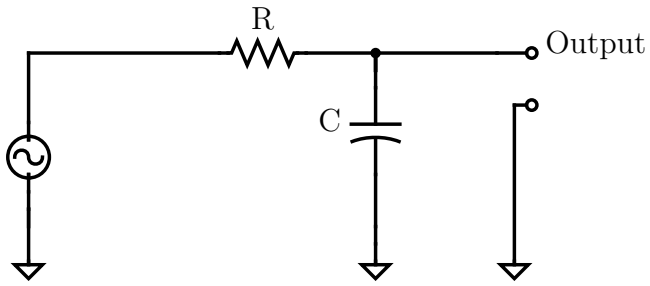


Figure 1: 1st Order Passive Low-Pass Filter

5.2 High-Pass Filter (Passive)

- Cutoff frequency: $f_L = 4[(G-1) \bmod 4] + 4$ kHz
  $f_L = 4[(6-1) \bmod 4] + 4$ kHz
  $f_L = 4[5 \bmod 4] + 4$ kHz
  $f_L = 4 * 1 + 4$ kHz
  $f_L = 8$ kHz

- Calculation:
  We have to taken $C = 9.42$ nF , Closest value of $R = 1.95$ kΩ , Corresponding Cutoff frequency 8.6 kHz .

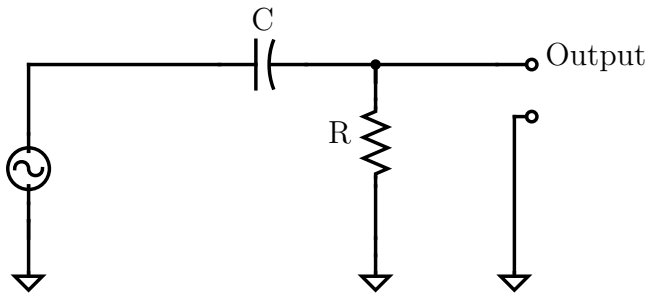


Figure 2: 1st Order Passive High-Pass Filter

5.3 Band-Pass Filter

- Cutoff frequencies: $f_L = 8$ kHz and $f_H = 14$ kHz

- $C1 = 8.36$ nF , $R1 = 1.18$ kΩ , $C2 = 9.42$ nF , $R2 = 1.95$ kΩ . Corresponding Cutoff frequencies 8.6 kHz and 16.12 kHz.

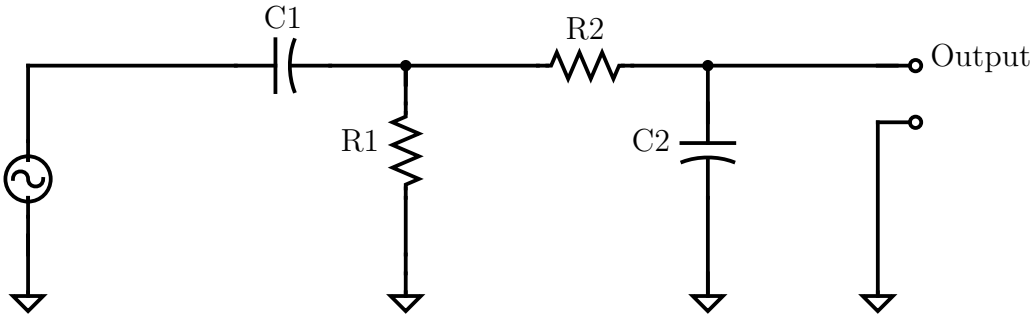


Figure 3: Band-Pass Filter using LPF and HPF

5.4 Active Low-Pass Filter

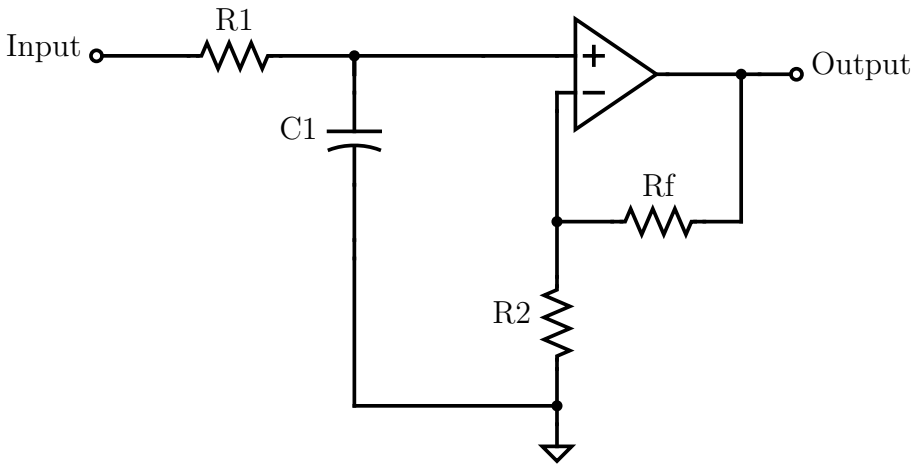


Figure 4: 1st Order Active Low-Pass Filter with Op-Amp

- Calculation: $Gain = 1 + \frac{Rf}{R2}$
  $Gain = 2$
  $=> 1 + \frac{Rf}{R2} = 2$
  $=> \frac{Rf}{R2} = 1$
  $=> Rf = R2$

  $C1 = 8.36$ nF , $R1 = 1.18$ kΩ , $Rf = R2 = 6.76$ kΩ

5.5 Adder Circuit

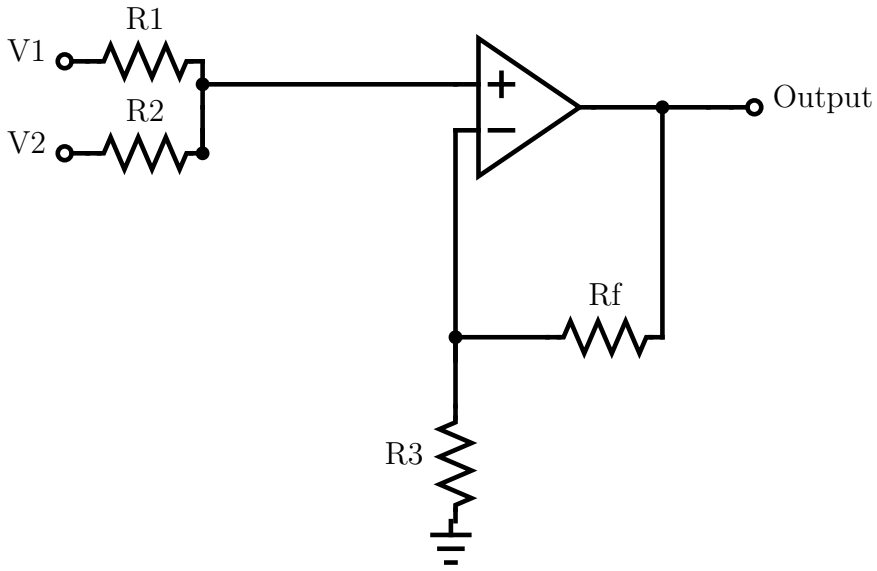


Figure 5: Adder Circuit using Op-Amp

$V_{\text{out}} = (-1)^G \left[x_1(t) + \big((G-1) \bmod 4 + 1\big)x_2(t)\right]$
$V_{\text{out}} = (-1)^6 \left[x_1(t) + \big((6-1) \bmod 4 + 1\big)x_2(t)\right]$
$V_{\text{out}} = \left[x_1(t) + \big(5 \bmod 4 + 1\big)x_2(t)\right]$
$V_{\text{out}} = \left[x_1(t) + \big(1 + 1\big)x_2(t)\right]$
$V_{\text{out}} = x_1(t) + 2x_2(t)$

- Calculations:
  Taking $R_f = R_3$ and By applying Ideal Op-Amp properties , we get $R_1 = 2 \times R_2$.
  Final Values: $R_f = R_3 = 6.8\text{k}\Omega$, $R_1 = 6.86\text{k}\Omega$ and $R_2 = 3.27\text{k}\Omega$.

6 Results and Discussion

- For Low Pass Filter due to unavailability of exact Resistances and Capacitance values , our cut-off frequency = 16.12 kHz. 3dB frequency was equal to 15.3 kHz.



Figure 6: Output Waveform of LPF at freq = 1 kHz



Figure 7: Output Waveform of LPF at freq = 14 kHz

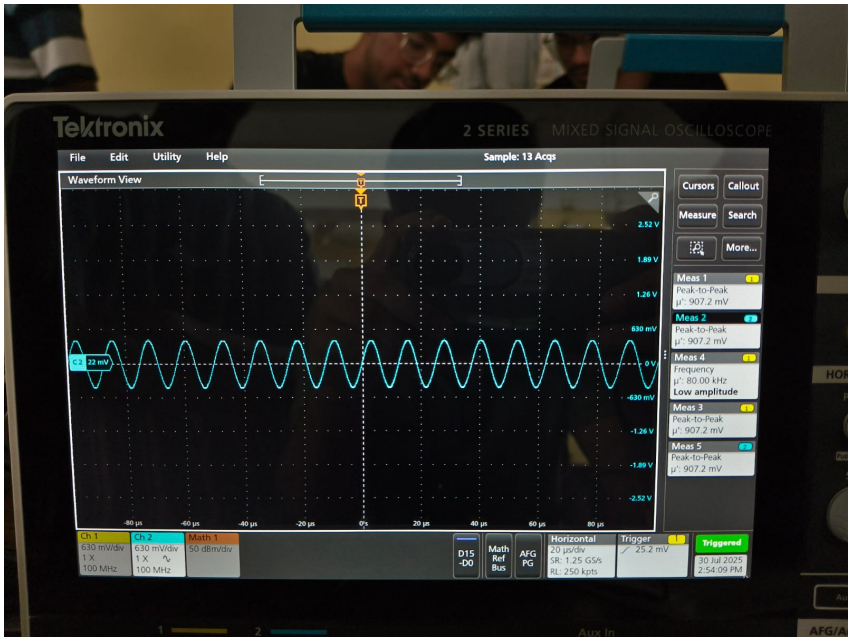


Figure 8: Output Waveform of LPF at freq = 80 kHz

- For High Pass Filter due to unavailability of exact Resistances and Capacitance values, our cut-off frequency = 8.6 kHz. 3dB frequency was equal to 7.2 kHz.

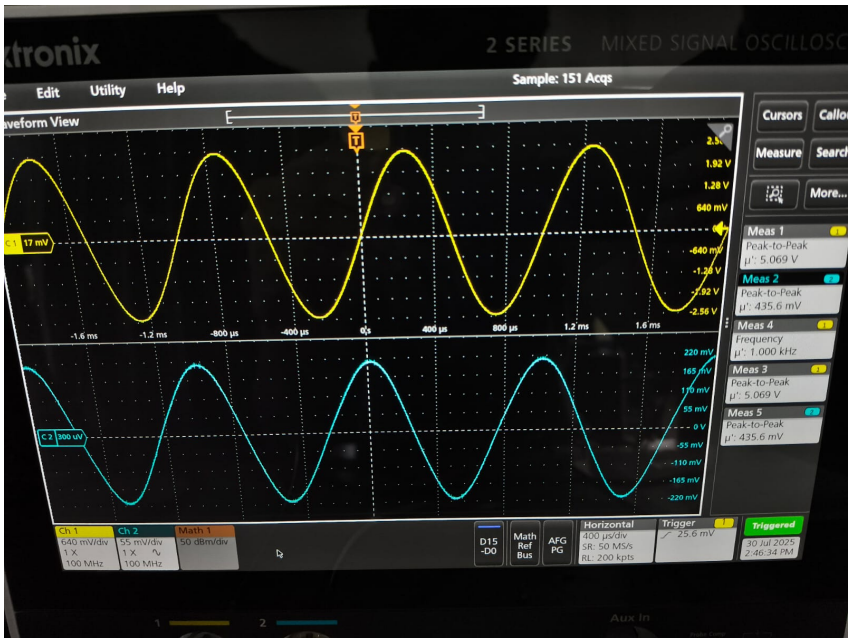


Figure 9: Output Waveform of HPF at freq = 1 kHz

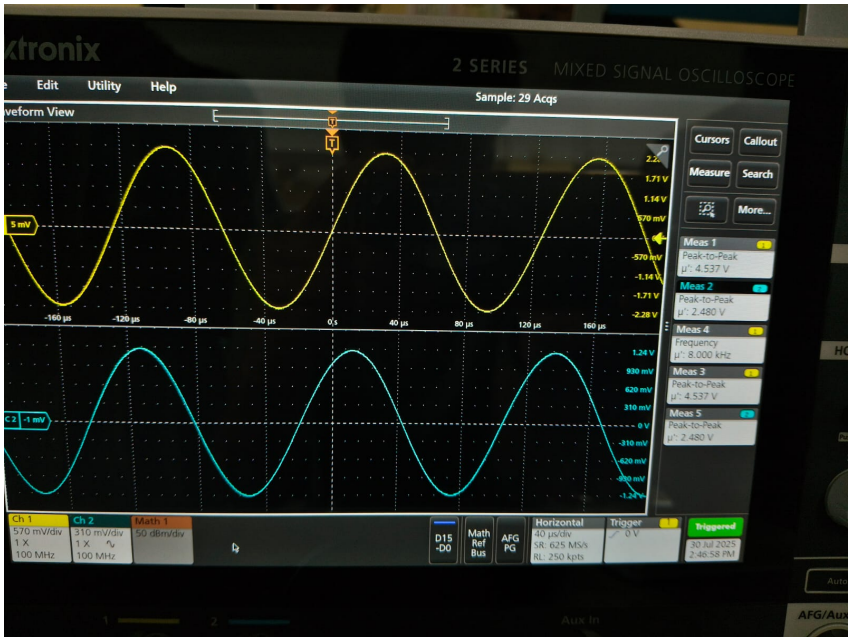


Figure 10: Output Waveform of HPF at freq = 8 kHz

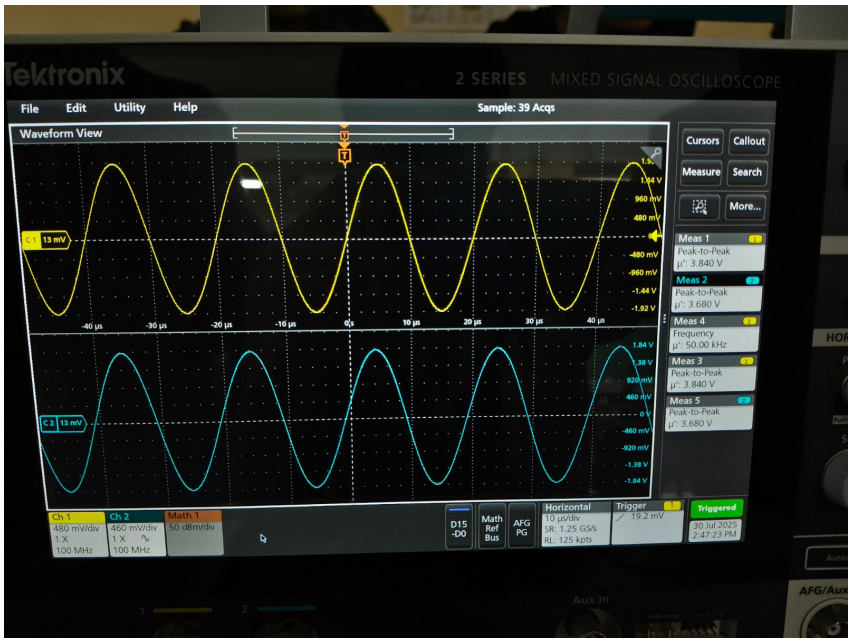


Figure 11: Output Waveform of HPF at freq = 50 kHz

- For Band Pass Filter due to loading of low pass circuit,Gain is less than expected, so we added a Op-Amp with Gain =1 between High Pass Filter and Low Pass Filter. it Increased Output Voltages significatly.

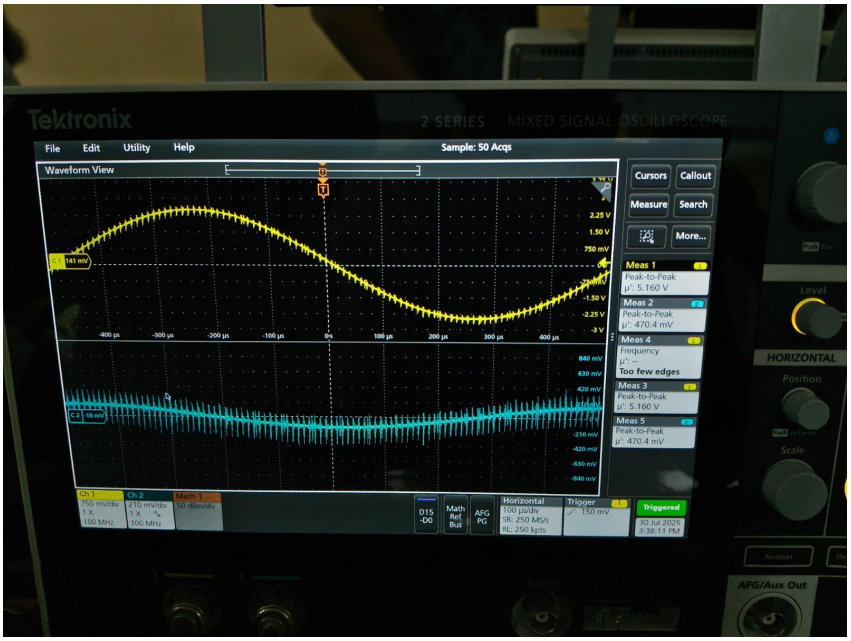


Figure 12: Output Waveform of BPF at freq = 1 kHz



Figure 13: Output Waveform of BPF at freq = 15 kHz

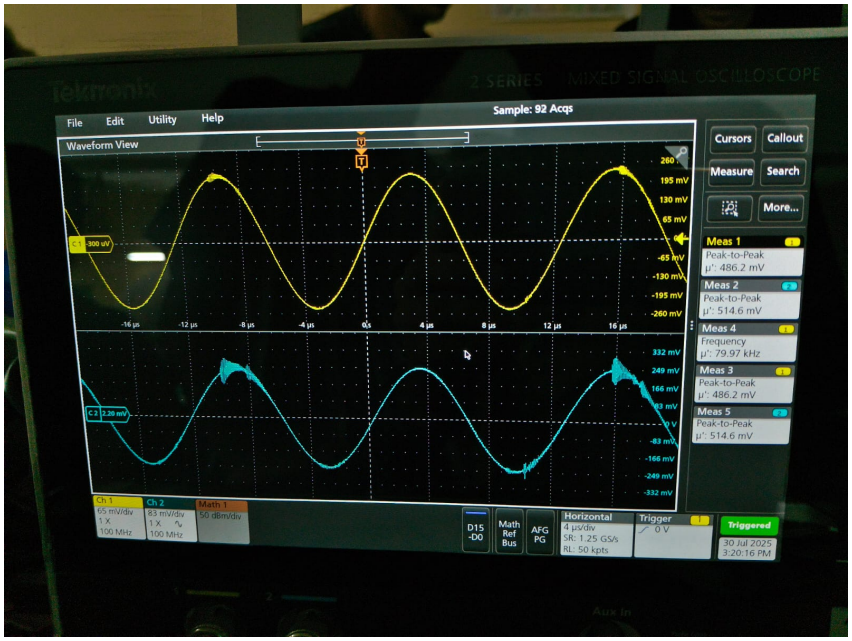


Figure 14: Output Waveform of BPF at freq = 90 kHz

- For Active Low Pass Filter ,we added an Op-Amp at the output stage which apmlifies the signal ,Since Gain is 2 . Output value increased as compared to passive low-pass filter.

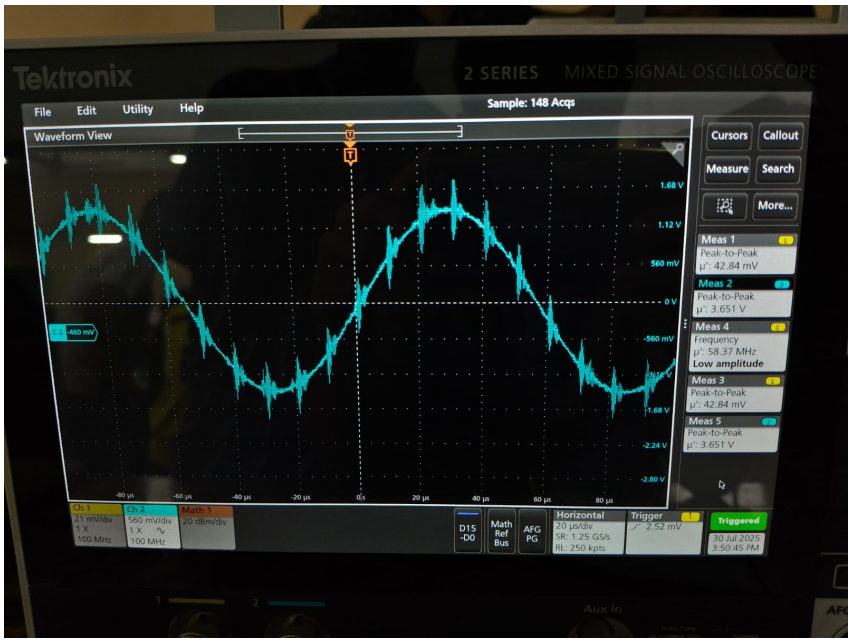


Figure 15: Output Waveform of active LPF at lower frequency near cutoff

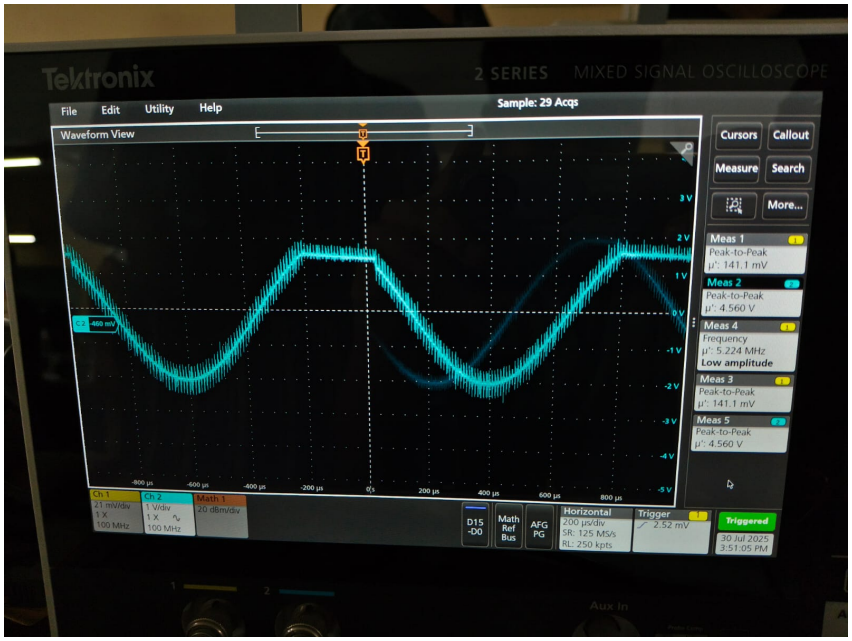


Figure 16: Output Waveform of active LPF at lower frequency

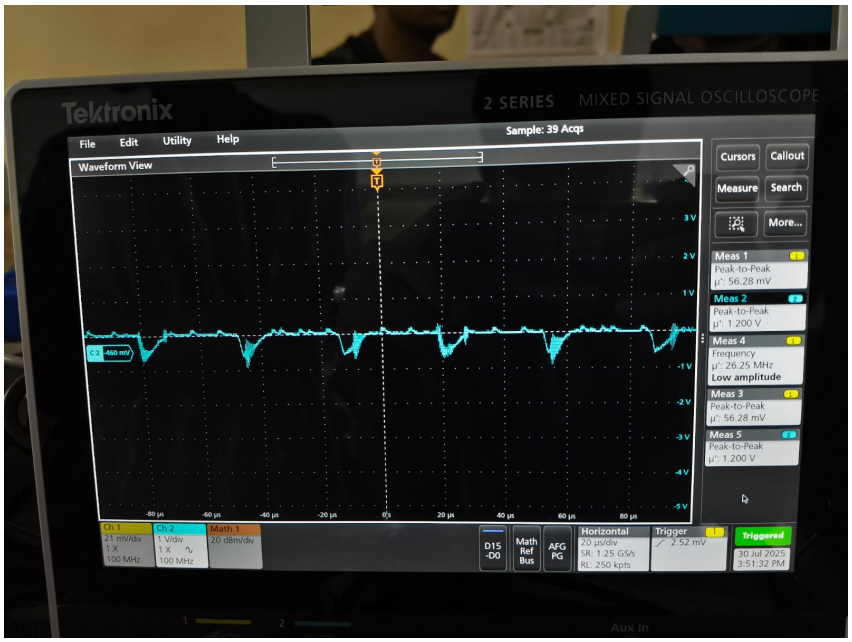


Figure 17: Output Waveform of active LPF at frequency greater than cut-off

- For Adder circuit our equation was $V_{\text{out}} = x_1(t) + 2x_2(t)$. Result were very close to expected value.

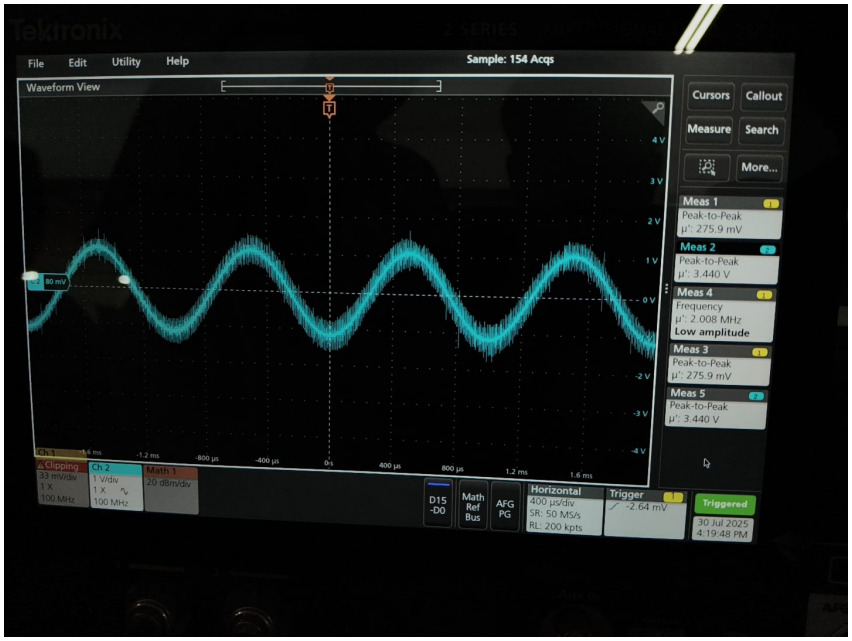


Figure 18: Output Waveform of Adder with $x_1 = 1V$ and $x_2 = 1V$,Output $= 3.44V$

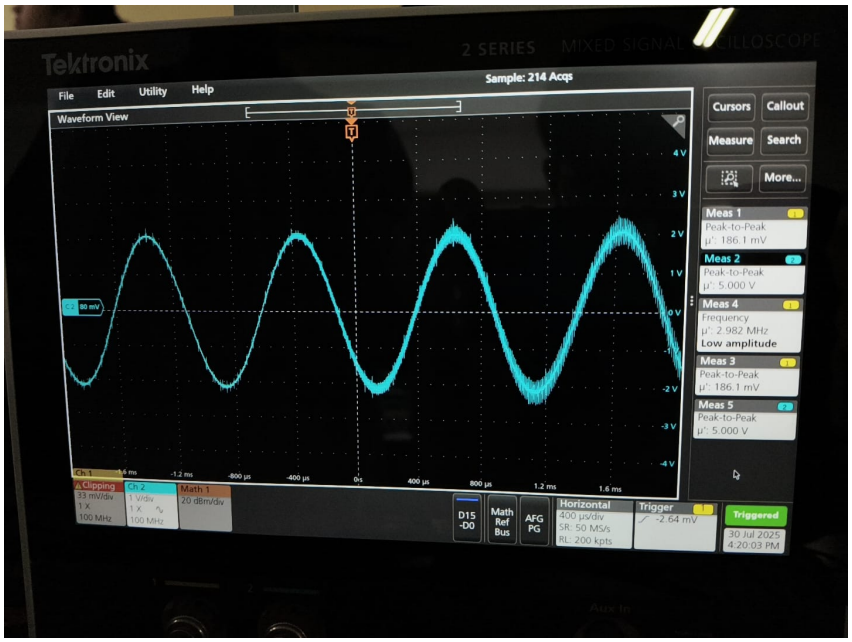


Figure 19: Output Waveform of Adder with $x_1 = 1V$ and $x_2 = 2V$,Output $= 5V$

- Measured 3dB frequencies matched closely with theoretical values.
- The adder circuit successfully added the input signals with correct polarity based on G.
- Observations confirmed the expected filter characteristics and Op-Amp behavior.
- The measured cutoff frequencies of the LPF and HPF closely matched the theoretical values, with only small deviations due to component tolerances.

- The BPF successfully passed signals between f_L and f_H while attenuating frequencies outside this range, validating the combined filter design.
- The active LPF provided amplification and minimized loading effects, producing a cleaner output compared to the passive LPF.
- Phase shifts observed in all filters (LPF, HPF, and BPF) aligned with theoretical predictions, confirming the accuracy of the design.
- The adder circuit accurately summed the input signals with the expected polarity and scaling factor, with only minor offsets caused by op-amp limitations.
- Overall, the experimental results were consistent with simulations and theoretical calculations, demonstrating proper filter and op-amp circuit functionality.

——END OF REPORT——